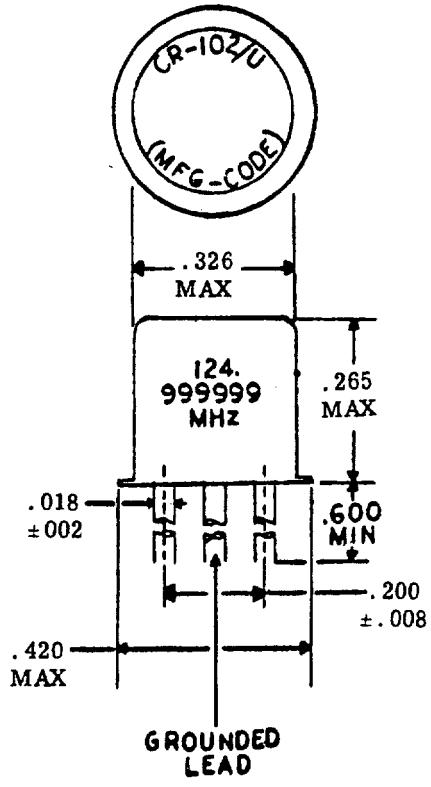
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-102/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the crystal unit described herein shall consist of this document and the latest issue of specification MIL-C-3098.

Pertinent characteristics: 50 to 125 MHz; fifth overtone; noncontrolled; series.



INCH	MM
.002	.05
.008	.20
.018	.46
.200	5.08
.265	6.73
.326	8.28
.420	10.67
.600	15.24

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. The crystal unit shall not be trimmed after sealing.

FIGURE 1. Dimensions and configuration.

Ⓑ denotes changes.

FSC 5955

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TABLE I. Requirements.

	Requirement	Specified value or condition
	Holder	HC-35/U (see MIL-H-10056/21)
	Dimensions and configuration	See figure
	Frequency range	50 to 125 MHz, inclusive
Ⓑ	Frequency tolerance	±25 PPM
	Capacitance, shunt	7 pF, maximum
Ⓑ	Equivalent resistance	50 to 90 MHz, 50 ohms, maximum 90 to 125 MHz, 60 ohms, maximum
	Mode of oscillation	Fifth overtone
	Operating temperature range	-55° to +105° C, inclusive
	Government reference test set	TS-683/TSM
	Calibration values:	
	Resistance	60 ohms
	Resistor voltage drop	0.35 volt
	Drive level, rated	2.0 ±0.4 mW
	Shock:	
Ⓑ	Frequency change permitted	±5 PPM
	Equivalent-resistance change permitted	±10 percent
	Vibration: 1/	
Ⓑ	Frequency change permitted	±5 PPM
	Equivalent-resistance change permitted	±10 percent
Ⓑ	Thermal shock:	
	Frequency change permitted	±5 PPM
	Equivalent-resistance change permitted	±10 percent
	Aging:	
Ⓑ	Frequency change permitted	±5 PPM

1/ Method 204, MIL-STD-202, test condition B.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI, AR, SM
Navy - SH
Air Force - 11, 99
DLA - ES

User activities:
Army -
Navy - AS, CG, MC, OS
Air Force - 17, 19

Preparing activity:
Army - EL

Agent:
DLA - ES

(Project 5955-0512-23)

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DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-C-3098/104B CRYSTAL UNIT, QUARTZ, CR-102/U

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

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